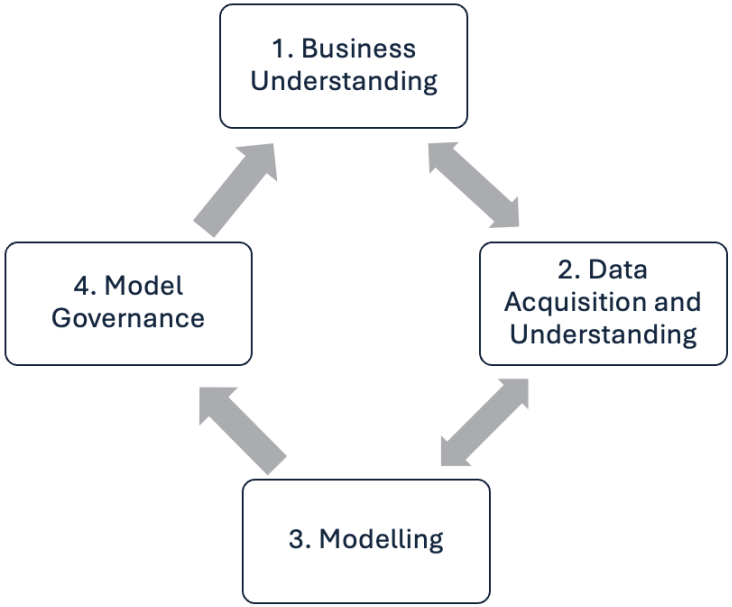
The Data Science Lifecycle

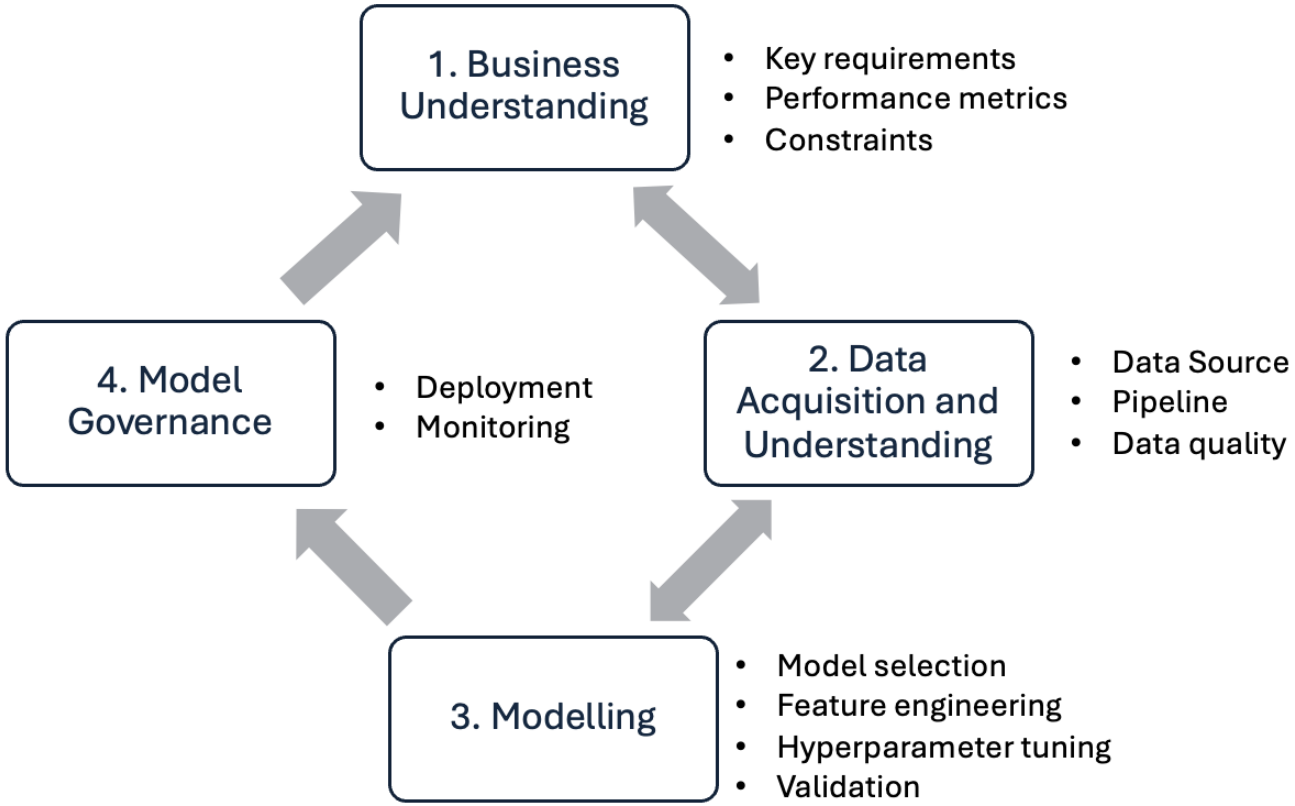


Learning goals

- Know phases of data science lifecycle
- Identify essential concepts, tools, and methods for each phase
- Plan an end-to-end ML workflow with attention to constraints and dependencies

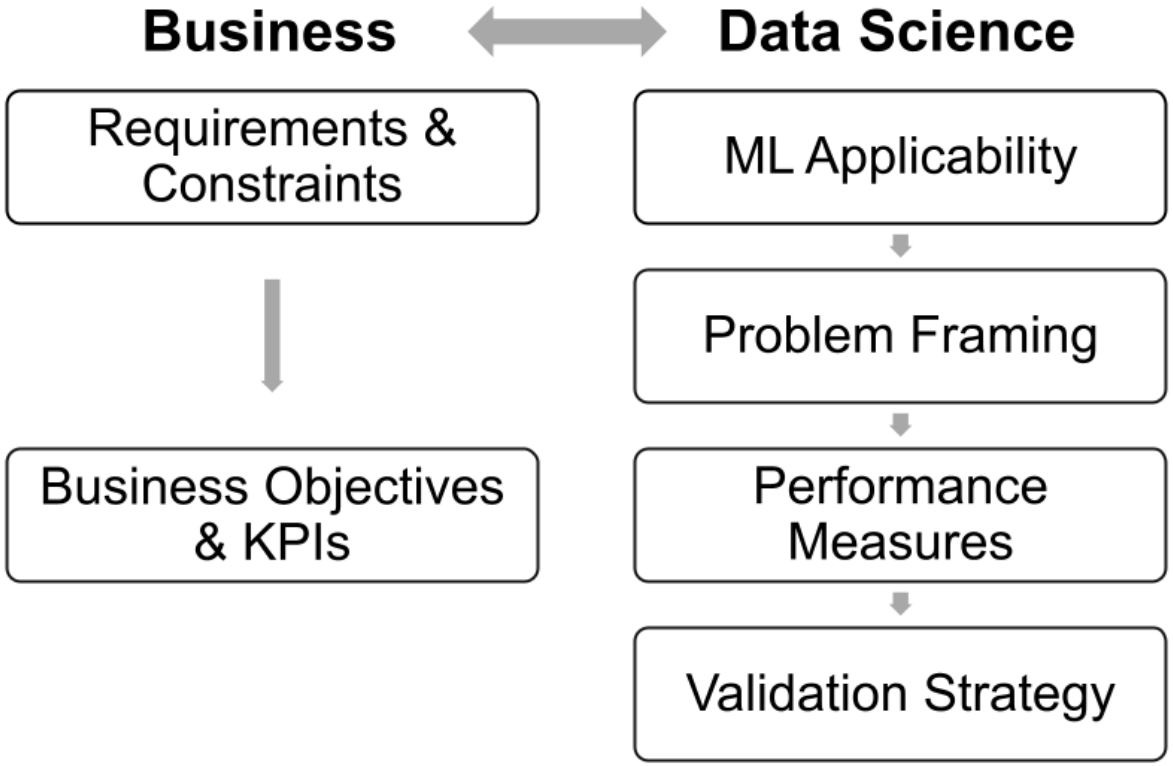
THE DATA SCIENCE LIFECYCLE ▸ CRISP-DM

Lifecycle Management: Process of managing a (product) lifecycle from inception, through engineering, design and manufacturing to deployment and eventual disposal.
CRISP-DM (Cross-Industry Standard Process for Data Mining): A widely used framework that formalizes the iterative process of turning data into knowledge.



1. Business Understanding

OVERVIEW BUSINESS UNDERSTANDING



REQUIREMENTS & CONSTRAINTS

Key question: What is the end goal of the project?

- Define the intended **output**:
  - Report, dashboard, or interactive tool?
  - Scalable pipeline for production use?

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Key question: What is the end goal of the project?

- Define the intended **output**:
  - Report, dashboard, or interactive tool?
  - Scalable pipeline for production use?
- Identify constraints
  - Ethical: Avoid sensitive data use (e.g., address may proxy ethnicity)
  - Legal: Exclude protected attributes (e.g., gender in automated hiring)
  - Reproducibility: Ensure the analysis is fully reproducible in a single step
  - Explainability: All model decisions must be explainable to the user
  - Prediction Latency: Real-time predictions required
  - Model Size: Stay within specified memory constraints

REQUIREMENTS & CONSTRAINTS

1. ML Applicability: Is ML justified?

- Explore non-ML baselines first (rules, heuristics, simple analytics)
- Use ML only if it adds clear value over simpler/existing alternatives

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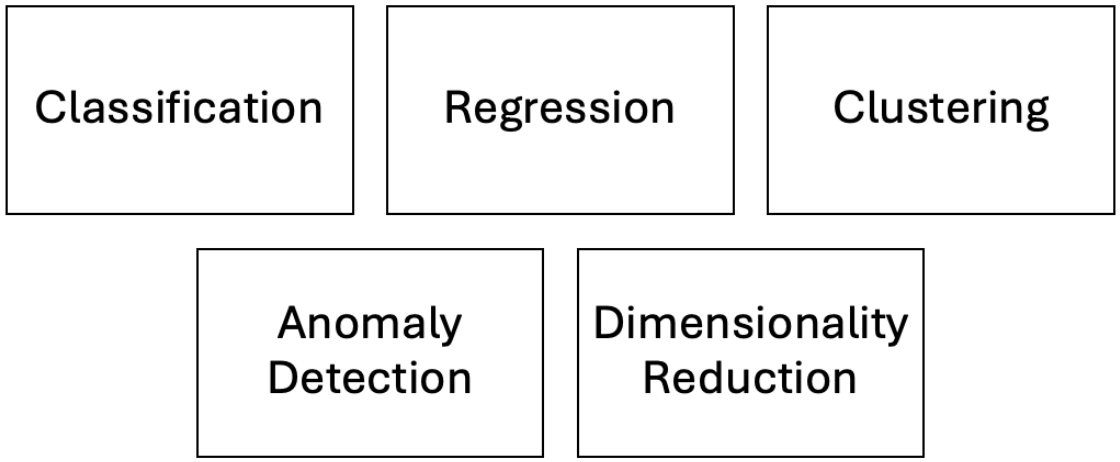
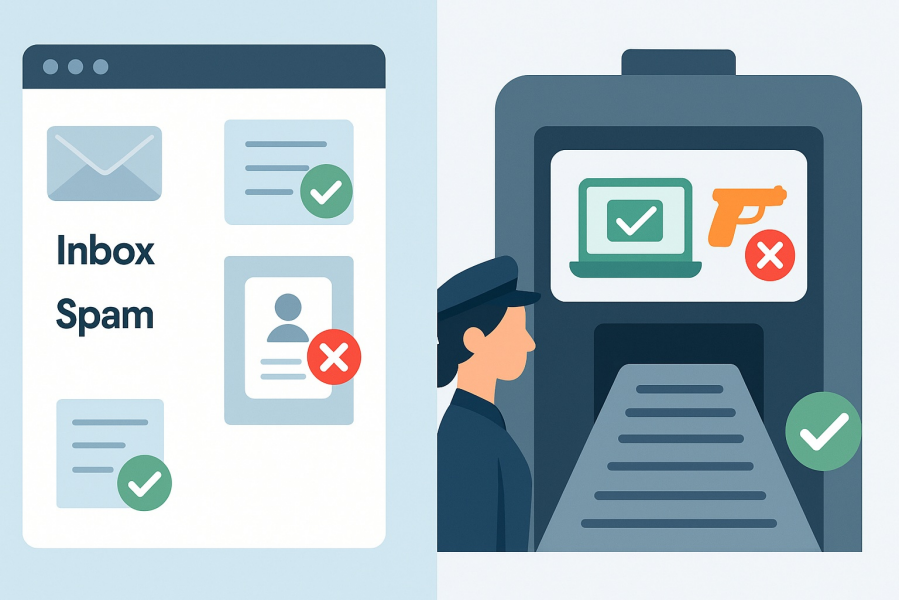
2. Problem Framing: How to frame the task?

- *Imbalanced data:* classification vs. anomaly detection?
- *Pred. maintenance:* failure probability (classif.) vs. remaining lifetime (regr.)?
- Select the ML task that aligns with your KPI and data characteristics



BUSINESS OBJECTIVES: PERFORMANCE MEASURES

- **How to translate key performance indicators (KPIs) to metrics?**
  - **Spam detection**: Job application flagged as spam = unacceptable
    $\Rightarrow$ Occasional inbox spam is fine $\Rightarrow$ minimize FP $\Rightarrow$ maximize **precision**
  - **Airport security**: Weapon flagged as laptop = unacceptable
    $\Rightarrow$ Occasional false alarm is fine $\Rightarrow$ minimize FN $\Rightarrow$ maximize **recall**



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- **Multiple objectives:** What if improving one metric worsens another?
  - *Scalarization*: Combine into one score (e.g., F_1)
  - *Pareto/frontier*: Analyze trade-off without collapsing (e.g., ROC curve)

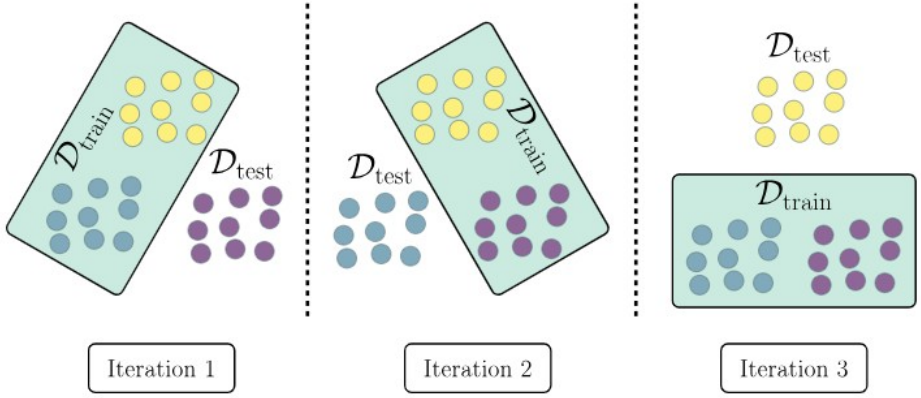
True Values	**Predicted Values**		
	Pos	**Neg**	
Pos	TP	FN	**Sensitivity/Recall/TPR** $= \dfrac{TP}{TP + FN}$
Neg	FP	TN	**Specificity** $= \dfrac{TN}{TN + FP}$
	Precision $= \dfrac{TP}{TP + FP}$	**NPV** $= \dfrac{TN}{TN + FN}$	$\mathbf{F}_1 = \dfrac{2\,\text{Precision Recall}}{\text{Precision} + \text{Recall}}$

BUSINESS OBJECTIVES: VALIDATION STRATEGY

Key question: How to validate the model reliably and realistically?

1. Strategic Considerations

- **What should your model generalize to?**
  - *Blocking factors* (e.g., customers, hospitals, product types)
  - *Temporal effects* (e.g., seasonality, trends)
  - *Spatial effects* (e.g., regions, branches)
- **Simulate realistic deployment:**
  - **Time series:** Train on data from 2016–2020, deploy in 2021
  - **Validation:** Mimic deployment – train on 2016–2019, validate on 2020



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2. Beware of Data Leakage

- **Target Leakage:** Features leak future info not available at prediction time
  - Example: "Total Sales in Next Month" used to predict this month
- **Train-Test Leakage:** Information from the test set is used during training
  - Example: Scaling the full dataset before splitting

2. Data Acquisition and Understanding

DATA SOURCES AND DATA VERSIONING

Key question: What data is available and where does it come from?

- **Internal vs. External:** External feeds may change format or be discontinued
- **Structured vs. Unstructured:** Tables vs. free-text, images, audio, etc.
- **Databases vs. Files:** DBs offer querying, consistency and security
- **On-Premise vs. Cloud:** Governed by regulations, cost and team expertise

Data Versioning:

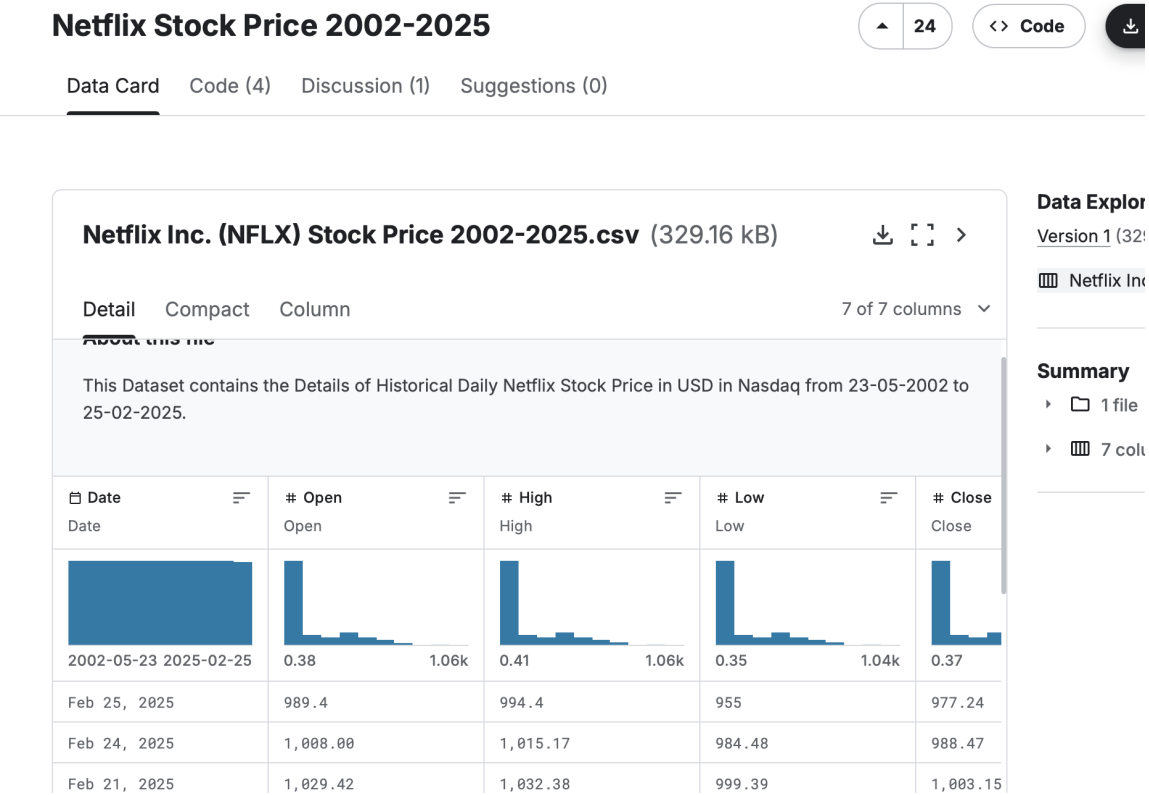
- Data can evolve - use version control for data, like `git` for code
- Tools: `DVC` (Data Version Control), `Pachyderm`

DATASHEETS

Gebru et al., 2018

Datasheets summarize key metadata about a dataset's content, structure, provenance, and known limitations to support informed and responsible use.

- Why and how was the data collected?
- What are its limitations and ethical concerns?
- For which tasks is it suitable or unsuitable?

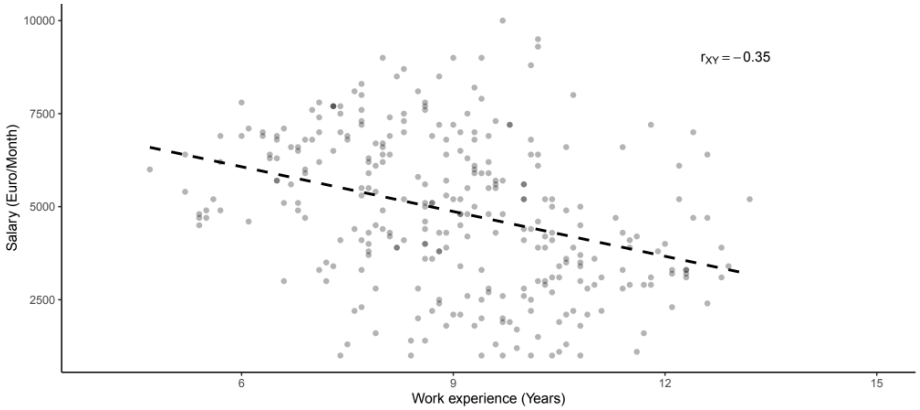


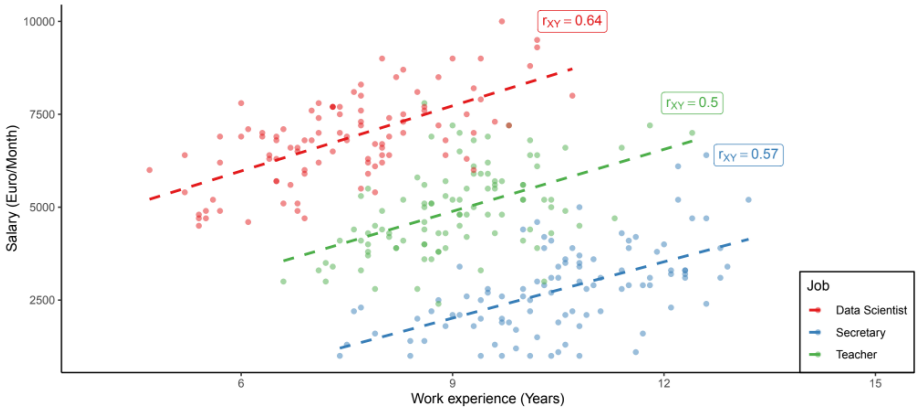
Source: Kaggle Datasets

DATA EXPLORATION

Key question: What features are in the data, and what do they reveal?

- **Build a data dictionary**:
  - Feature meaning, units, valid ranges/category-levels
  - Missing value encoding and special codes
- **Examine structure and quality to gain understanding**:
  - Compute (stratified) summary statistics and inspect missingness
  - Plot univariate and bivariate distributions
  - Identify outliers, anomalies, imbalances, redundancy, or high correlation
- **Watch for confounders:** unmeasured variables may distort associations





DATA QUALITY

Key question: Is the data fit for modeling? *Garbage in, garbage out.*

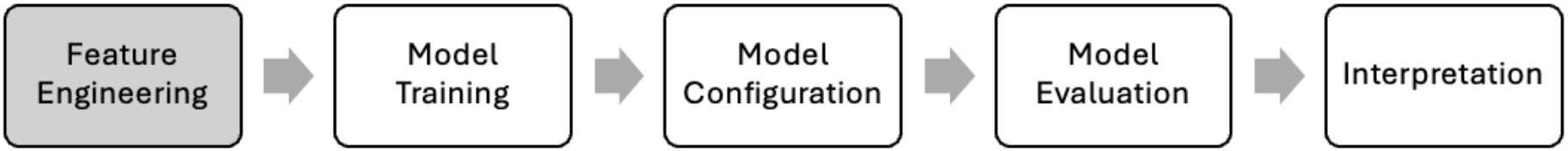
- Assess missingness: How many values are missing, and where?
- Detect implausible or inconsistent values (requires domain expertise)
- Standardize formats (e.g., date/time parsing)
- Drop near-constant or non-informative features
- Remove redundancy and obsolete variables

Watch for data bias:

- *Sampling bias:* E.g., a survey at 10am in the city center misses working adults
- *Response bias:* Online reviews often reflect extreme opinions, not the median

3. Modeling

FEATURE ENGINEERING



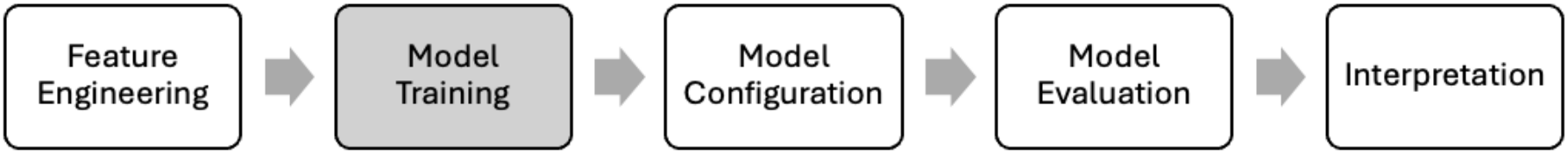
Feature Engineering transforms raw data into informative input variables for modeling.

- **Transformation**: Scaling, log-transform, one-hot encoding
- **Selection**: Wrapper and filter methods, model-based importance
- **Dimensionality Reduction**: PCA, t-SNE
- **Creation**: Aggregation, binning, interaction terms

Best Practices:

- Avoid overengineering: irrelevant features introduce noise
- Fit transformations **only on training data**; reuse parameters for test data
- Prevent leakage: never use test data in feature construction

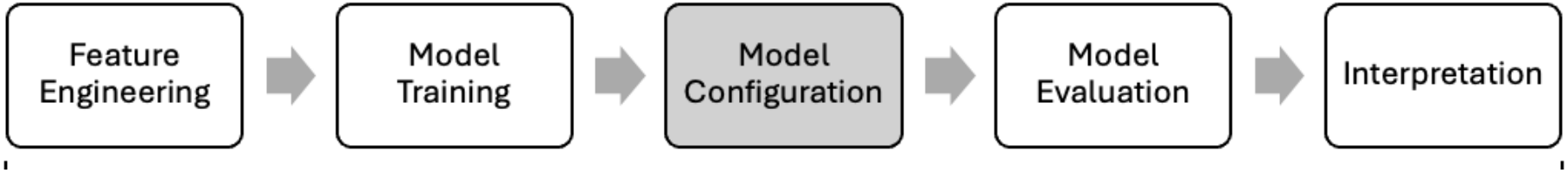
MODEL TRAINING



Treat model training like a scientific experiment:

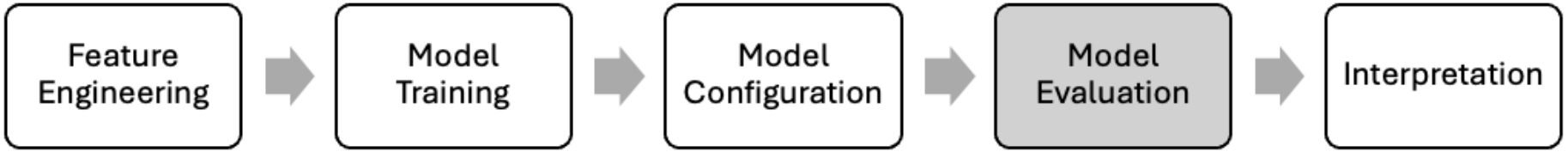
- **Track all inputs precisely:**
  - Dataset version
  - Algorithm and hyperparameters
  - Code version (e.g., Git commit/tag)
- **Proceed iteratively:**
  - *Start simple* – build a fast, interpretable baseline
  - Gradually increase model complexity and validate each improvement

MODEL CONFIGURATION



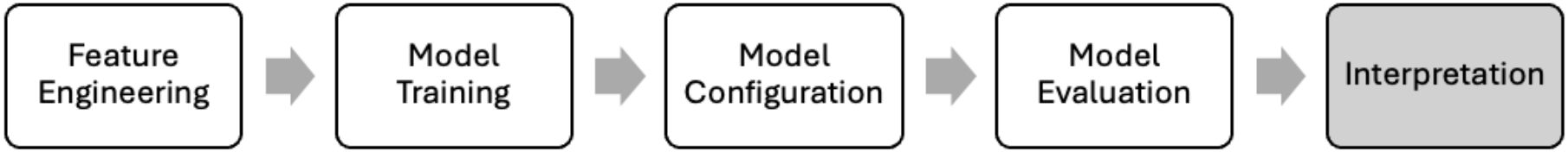
- How do you track the exact configuration of a model experiment?
- How do you record small changes between experiments?
- Store in plain text (e.g., `.yaml`, `.json`) for readability and version control
- Recommended tool: structured configuration managers
  - **Python**: Hydra - structured, hierarchical configuration management
  - **R**: config - environment-based YAML setup

MODEL EVALUATION



- Select models based on predefined **metrics** (e.g., accuracy, AUC, RMSE)
- Use the **validation or test set** in line with your validation strategy
- Assess **robustness**: How does the model respond to noise, perturbations, or adversarial inputs?
  $\Rightarrow$ Add synthetic noise and measure changes in model parameters/performance

INTERPRETATION AND FAIRNESS



To understand and trust your model, apply methods from **Interpretable Machine Learning (IML)**:

- **Feature Importance**: Which features drive predictions globally?
- **Effects & Interactions**: Visualize marginal effects and feature interactions
- **Local Explanations**: Explain individual predictions
- **Bias Detection**: Reveal fairness issues and unintended discrimination
  - Example: Amazon's 2018 hiring tool penalized resumes with "women's" due to biased training data ▸ The Guardian, 2018

MODEL CARDS FOR MODEL REPORTING

Mitchell et al. 2019

Model Card: Documentation of the model similar to a Datasheet for data

- **Model Details**: Algorithm, parameters, model version
- **Intended Use**: Intended use and out-of-scope use cases
- **Performance**: Key metrics, validation approaches
- **Analyses**: Feature/target distributions in train/test data
- **Ethical Considerations**: Failure modes, fairness concerns, inappropriate use

Model Card for Breast Cancer Wisconsin (Diagnostic) Dataset

Model Details

Overview

This model predicts whether breast cancer is benign or malignant based on image measurements.

Version

name: bba6bec9-9d5c-4f0e-a291-72125c2c534a
date: 2020-09-25

Owners

- Model Cards Team, model-cards@google.com

References

- https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)
- https://minds.wisconsin.edu/bitstream/handle/1793/59692/TR1131.pdf

Considerations

Intended Users

- Medical professionals
- ML researchers

Use Cases

- Breast cancer diagnosis

Limitations

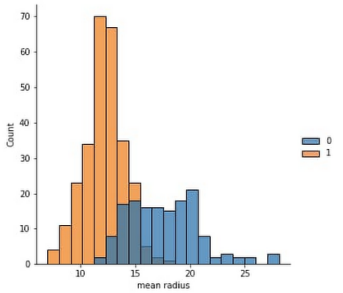
- Breast cancer diagnosis

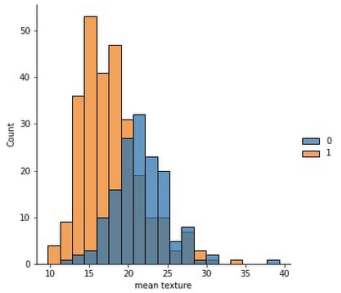
Ethical Considerations

- Risk: Manual selection of image sections to digitize could create selection bias
  Mitigation Strategy: Automate the selection process

Train Set

426 rows with 30 features





4. Model Governance

MODEL GOVERNANCE

- Model Governance includes **people, processes, and technologies** needed to manage and protect your ML models.
- Ensures long-term maintainability and adaptation as requirements evolve.
- Two extremes must be avoided:
  - **Repression**: too many rules, rigid standards, excessive control
  - **Chaos**: too much speed, freedom, creativity, and unstructured change

Model governance steps are:

"Developing and deploying ML systems is relatively fast and cheap, but maintaining them over time is difficult and expensive." *Hidden Technical Debt in ML Systems*

DEPLOYMENT

Deployment means making a model accessible to users (humans or other systems).

- Common deployment strategies:
  - **One-off**: Manually trained and deployed once; no regular updates
  - **Batch**: Periodically retrained on recent data (e.g., weekly updates)
  - **Real-time**: Continuously updated on streaming data; often latency-critical
- For frequent updates, establish model **versioning and lifecycle tracking**
  - Tools like MLflow provide version control and a model registry

MONITORING

Monitoring ensures that deployed models work reliably and flags issues in production:

- **Pipeline Failures**: Bugs in data ingestion or transformation logic
- **Data Drift**: Changes in input distribution
  - Example: An image classifier trained on summer data fails when images contain winter snow
  - Also affects evolving targets (e.g., new categories or value ranges)
  - **Remedy**: Label sufficient new data and retrain the model
- **Concept Drift**: Changes in the meaning of labels
  - Example: Redefining “high blood pressure” from 140mmHg to 135mmHg
  - **Remedy**: Relabel historical data and retrain

USER FEEDBACK, REINFORCEMENT

Handover: At project completion, engage users and transfer responsibilities:

- **System Validation**: Confirm with stakeholders that the solution meet their needs
- **Project Handover**: Transfer system ownership to the production team

Feedback loops occur when model outputs influence future inputs:

- **Direct Loops**: A model affects the data it later learns from
  - Example: Netflix recommends certain movies, then only learns from user responses to those
- **Hidden Loops**: Multiple models indirectly affect each other via the environment
  - Example: Flash crashes triggered by interacting trading algorithms amplifying market reactions

Best Practices for Competition Success

STEP 1: UNDERSTAND THE PROBLEM STATEMENT

- **Clarify Objectives & Business Impact**: Determine whether it's a classification or regression task, the domain context, and what "success" looks like.
- **Target Variable**: Understand how the target is defined and distributed.
- **Constraints & Permissions**: External data limits, privacy, or regulatory issues.
- **Action Points**:
  - Annotate key requirements and assumptions.
  - Research the domain for feature ideas.
  - Restate the problem in your own words to ensure clarity.

STEP 2: STUDY THE EVALUATION METRIC

- **Metric Matters**: Each competition or project has a specific metric (e.g., F1-score, AUC-ROC, RMSE), which guides how you tune and compare models.
- **Classification vs. Regression Metrics**:
  - Accuracy, F1-score, AUC-ROC, and Log Loss for classification.
  - RMSE, MAE, R^2 for regression.
- **Optimization Direction**: Decide if you must maximize or minimize the metric; some metrics (e.g., AUC-ROC) may need a proxy loss for training.
- **Action Points**:
  - Simulate prediction changes to see impact on the metric.
  - If allowed, implement custom losses aligned to the competition metric.

STEP 3: PERFORM AN INITIAL DATA ANALYSIS (EDA)

- **Data Profiling**: Summarize shape, basic statistics, feature and target distribution.
- **Quality Checks**: Identify missing values, outliers, or suspicious patterns; confirm no hidden data leakage.
- **Train-Test Shift**: Check if train and test set differ significantly in their feature distributions.
- **Action Points**:
  - Use tools like Pandas Profiling or Sweetviz for automated EDA.
  - Document anomalies or domain-specific oddities.
  - Examine correlations, scatter plots, and histograms.

STEP 4: DEVELOP A BASELINE MODEL

- **Why a Baseline?**
  - A simple model (e.g., LM, Decision Tree) provides a reference performance.
  - Include a *featureless* model (e.g., predicting the majority class or mean) to benchmark against trivial solutions.
- **Pipeline Check**: Make sure that data loading, preprocessing, and validation splits are correct and leakage-free.
- **Diagnostics** : Analyze misclassifications/residuals to spot improvement areas.
- **Action Points**:
  - Start with a naive or default-parameter model.
  - Use cross-validation to assess generalization.
  - Log baseline metrics to quantify future gains.

STEP 5: SET UP A ROBUST VALIDATION STRATEGY

- **Importance of Validation**: Avoid overfitting or a single random split.
- **Common Methods**:
  - K-Fold (balanced data).
  - Stratified K-Fold (imbalanced classes).
  - Time-Based or Group Splits (time series, repeated measures).
- **Action Points**:
  - Match validation approach to the final test scenario.
  - Keep a holdout set for final checks.
  - Use consistent folds across experiments to compare fairly.

STEP 6: FEATURE ENGINEERING & PREPROCESSING

- **Missing Data**: Mean/median imputation, advanced methods, or flags.
- **Encoding Categorical Variables**: One-hot/ target/ frequency enc. (leakage?!).
- **Scaling & Transformations**: Normalize or log-transform skewed features, especially for sensitive models (SVMs, neural nets).
- **New Features**: Time-based, domain knowledge, interactions (e.g. ratios).
- **Action Points**:
  - Add features in small batches and track validation changes.
  - Use scikit-learn Pipelines to avoid leakage.
  - Consider dimensionality reduction if feature space is large.

STEP 7: SELECT & TUNE MODELS STRATEGICALLY

- **Model Choices**:
  - Tree-based ensembles (XGBoost, LightGBM, CatBoost) for tabular data.
  - Neural networks for large-scale or unstructured data.
  - Simple linear/logistic models can still compete with good feature engineering.
- **Hyperparameter Tuning**:
  - Grid Search, Random Search, Bayesian Optimization.
  - Early stopping and regularization (L1, L2) to avoid overfitting.
- **Action Points**:
  - Track experiments (MLflow, Weights & Biases, or spreadsheets).
  - Start with defaults and refine gradually.
  - Use IML tools (SHAP, permutation importance) for feature insights.

STEP 8: TRACK EXPERIMENTS AND ITERATE

- **Experiment Logging**: Keep detailed records of data versions, hyperparams, code revisions.
- **Error Analysis**: Study misclassifications and residuals to guide feature tweaks.
- **Iterative Improvements**:
  - Refine features, adjust validation, revisit transformations.
  - Maintain a “changelog” for each iteration.
- **Action Points**:
  - Automate repeated tasks (submissions, data prep).
  - Prune ineffective ideas swiftly.
  - Ensure reproducibility (version control, environment snapshots).

STEP 9: UNDERSTAND THE LEADERBOARD

- **Public vs. Private Leaderboard**: Public boards use only part of the test set, overfitting leads to dramatic rank changes later.
- **Submission Strategy**:
  - Rely mainly on internal validation.
  - Keep a private holdout to confirm final performance.
- **Leaderboard Shakeups**: Watch for big changes when private scores are revealed; it often indicates overfitting to the public set.
- **Action Points**:
  - Limit minor tweaks solely to improve a public rank.
  - If a jump seems suspicious, re-check data handling for leakage.

STEP 10: COLLABORATE AND LEARN FROM OTHERS

- **Study Top Solutions**: Many winning approaches are documented in blog posts or Kaggle discussions.
- **Teamwork**:
  - Collaboration brings diverse perspectives and model ensembling.
  - Share code and features for synergy.
- **Community & Networking**:
  - Discussion forums reveal dataset quirks, best practices.
  - Local meetups or online communities keep you updated.
- **Action Points**:
  - Document and share your findings.
  - Adapt others' ideas carefully—validate them on your own splits.
  - Engage with peers for new tools and techniques.

FINAL ADVICE

- **Start Simple, Iterate**: Build a solid baseline before adding complexity.
- **Prioritize Validation**: A robust validation strategy prevents nasty surprises.
- **Log Everything**: Document transformations, parameters, and model versions.
- **Aim for Generalizable Solutions**: Don't just chase leaderboard scores, keep your focus on reproducibility and reliability.